

Literature Status for the WASQ and SSD2P Questions in arXiv:2205.02685

Run fa_banach_001, lane 1

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Status

This note records that two questions from Haller–Kaasik–Ostrak, arXiv:2205.02685, are answered negatively by Kaasik–Veeorg, arXiv:2210.09158.

No new theorem is claimed here. The purpose of this packet is to identify the exact later results that settle the source-paper questions.

Source Questions

The source paper proves that, for complete metric spaces, $\mathcal{F}(M)$ is locally almost square exactly in the length-space case, and that no Lipschitz-free space is almost square. Immediately after these results are announced, the authors ask whether $\mathcal{F}(M)$ is weakly almost square whenever M is length, and whether any Lipschitz-free space can have the symmetric strong diameter 2 property.

Later Answers

The later paper arXiv:2210.09158 explicitly cites arXiv:2205.02685 as HKO. Its abstract already states the two relevant conclusions: it constructs a Lipschitz-free space which is LASQ but not WASQ, and it proves that a Lipschitz-free space cannot have SSD2P.

For the WASQ question, Section 2 defines

$$M = [0, 1] \times (0, 1/2] \cup \{(0, 0), (1, 0)\}$$

with a metric that moves horizontally within each level and otherwise through one of the two endpoints. The paper observes that M is a length space, so $\mathcal{F}(M)$ is LASQ by arXiv:2205.02685. Theorem **notWASQ** then proves that $\mathcal{F}(M)$ is not WASQ. Therefore the length-space condition does not imply WASQ for Lipschitz-free spaces.

For the SSD2P question, Section 4 proves Theorem **thm: F(M) never SSD2P**: a Lipschitz-free space cannot have SSD2P. The theorem is stronger, excluding SSD($2d$)P for every $0 < d \leq 1$.

Scope

The two explicit questions from arXiv:2205.02685 are settled negatively in the later paper. The later paper leaves a refined problem open: whether $\mathcal{F}(M)$ is WASQ exactly when M is geodesic. It also proves a finite-support positive result for geodesic spaces, but that is not needed for the two negative answers recorded here.

References

1. R. Haller, J. K. Kaasik, and A. Ostrak, *The Lipschitz-free space over length space is locally almost square but never almost square*, arXiv:2205.02685.
2. J. K. Kaasik and T. Veeorg, *Weakly almost square Lipschitz-free spaces*, arXiv:2210.09158.